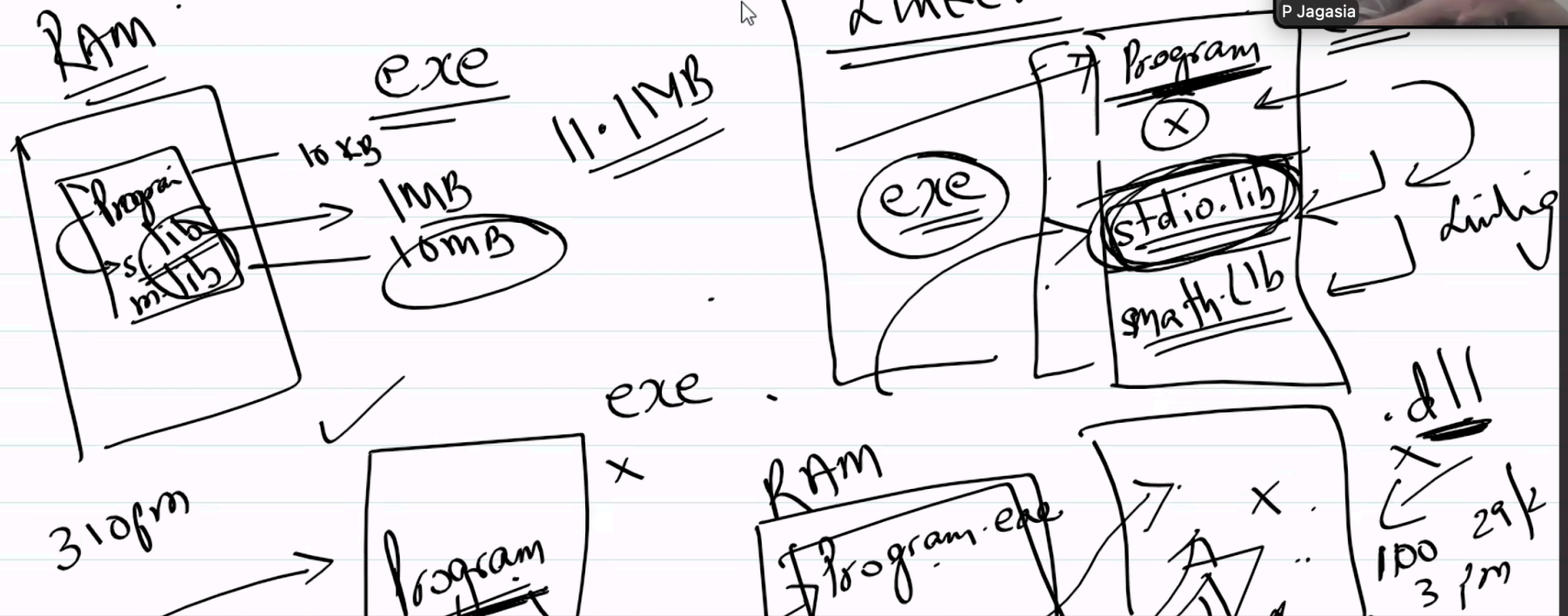




- ① SLL } → static linked library  
② DLL } → Dynamic



P Jagasia



Search

29°C  
Mostly cloudyENG  
IN15:42  
08-09-2026

① Procedural → C

② Object Based → V Basic

→ ③ Object Oriented Language → Java C++  
.NET,

→ Object Real time Entity

a) Identity

b) State or Properties

c) Behaviours / Methods

C → process-h  
-beginthread ex

ADP.



OOP.

Major

- ① Abstraction
- ② Encapsulation / Hiding
- ③ Inheritance
- ④ Polymorphism
- ⑤ Modularity

Minor

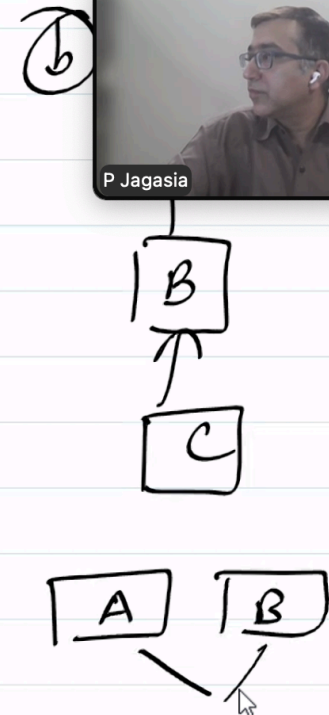
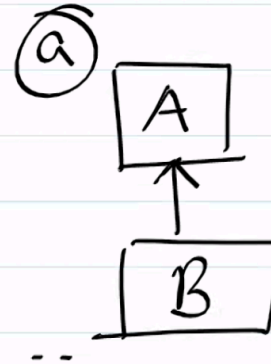
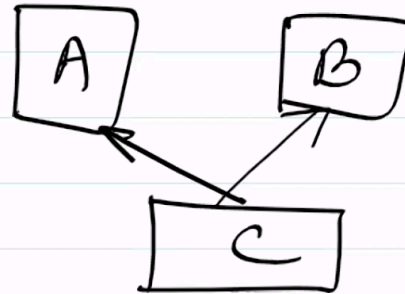
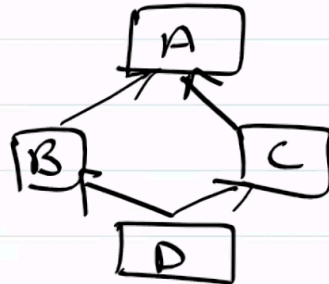
- ⑥ Persistence / Serialization
- ⑦ Concurrency
- ⑧ RTTI / RTCI  
Runtime Type Identification  
or  
Runtime Class Information



## Runtime Class Information

### ③ Inheritance

- a) Single.
- b) Multi level
- c) Multiple
- d) Hybrid





class A

```
{  
    run  
    int data;  
    void display()  
    {  
    }  
}
```

};

```
class C: public A, B  
{  
    ?  
}
```

class B

```
{  
    run  
    int data;  
    int num;  
    void show()  
    {  
    }  
    void display()  
    {  
    }  
}
```

}

3;

3

16 bytes obj 200

C obj;

obj.data = 5;

obj.A::data = 5;

obj.B::data = 6;

obj.num = 7;

obj.value = 8;



A  
203

204  
B 207  
208

211  
new members  
212  
215

(4)

## Polymorphism

- a) Compile / Static / Early binding
- b) Runtime / Dynamic / Late binding

### (a) Overloading

- a function with same name but
- i) either the type of para change or
  - ii) no. of para change
  - iii) sequence of datatype of para

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$$\sim (\underbrace{\%d \%d}_{\text{num}}, \text{num}, \text{num});$$

```
int printf(const char r, ...);
```

↑ ellipses  
variable argument  
list



int printf(const char \*, ...);

↑ ellipses  
variable argument  
list

Name mangling → Compiler

C

function name

printf(---);  
@printf

C++

function name  
+  
parameters

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ENG  
IN



76%

17:08  
08-09-2026





+ parameters



P Jagasia

(a) Monolithic Based Arch.

App 2 Lenovo  
3 Dell  
4 Intel  
5

⑥ Persistence / Serialization  
1, British, Java, 99999

⑦ Concurrency

Student

rollno =  
name =  
m o  
con =  
? British  
xxx 9999  
abc java

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Search



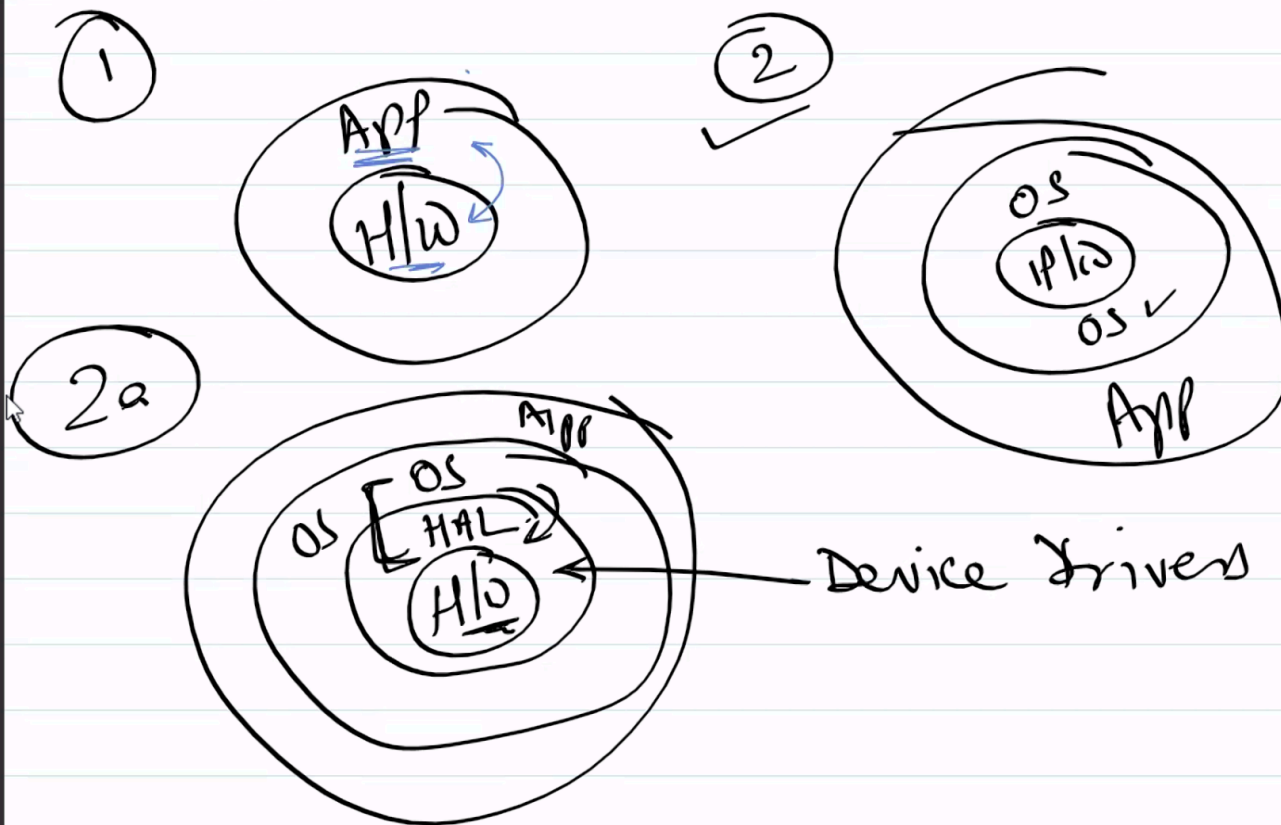
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Partly sunny

ENG  
IN



76%

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08-09-2026



Search



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Partly sunny



ENG  
IN



76%

18:18  
08-09-2026



COFF → Windows

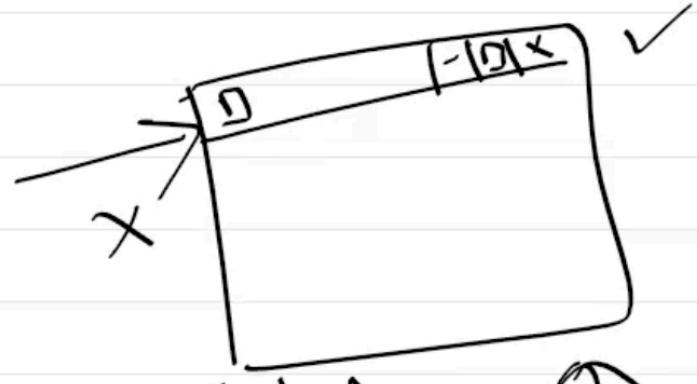
① Format of binary

② API

API  
OS

Windows ✓

CreateWindow (1)



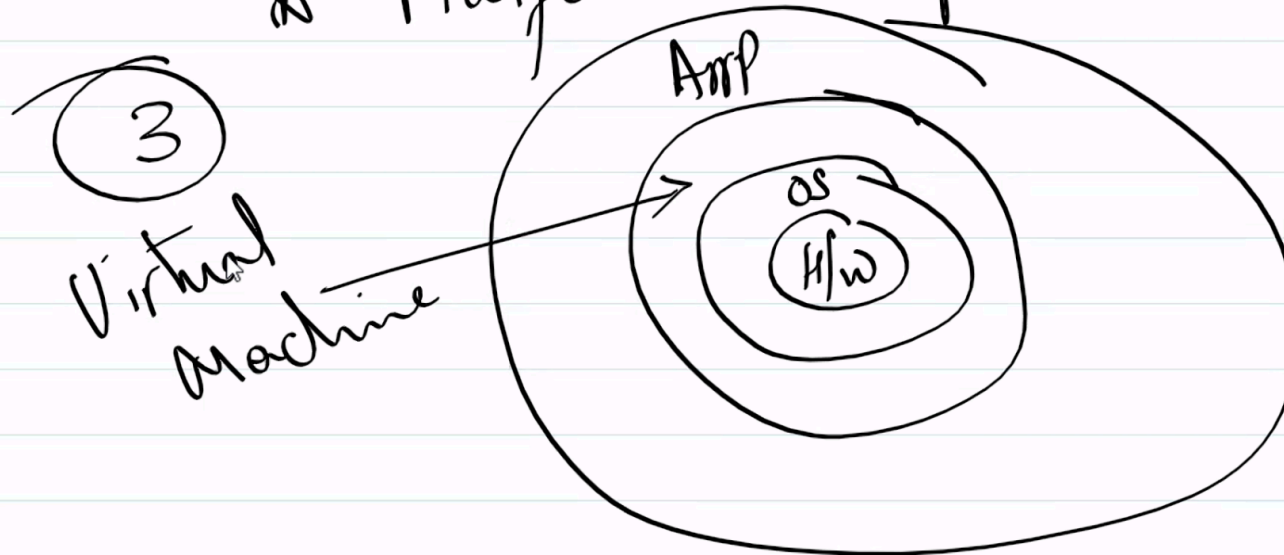
↓ Linux

OpenWindow (3);

James Gosling → Sun Micro

1990 → 2000 → 200 microprocessors

\* Platform Independent



P Jagasia



Search



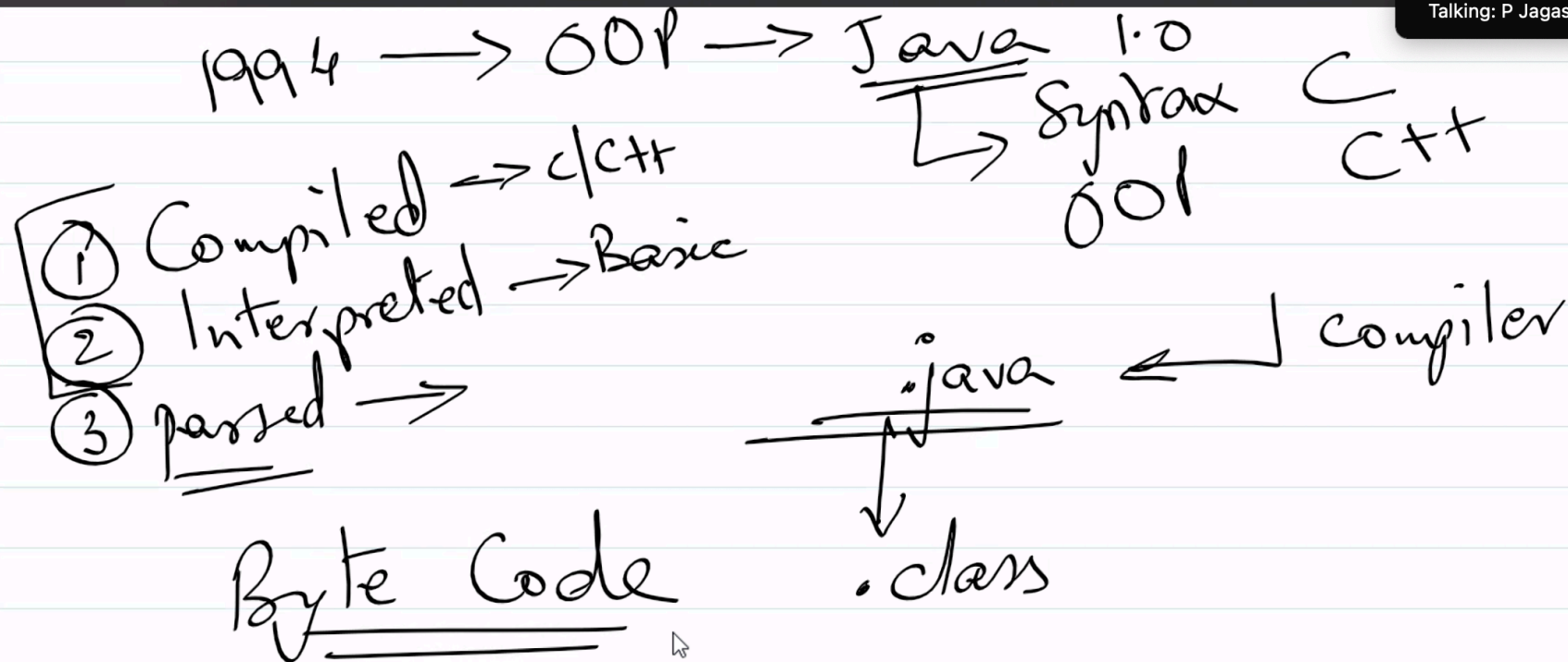
27°C  
Partly sunny



ENG  
IN



18:31  
08-09-2026



Search

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Partly sunnyENG  
IN

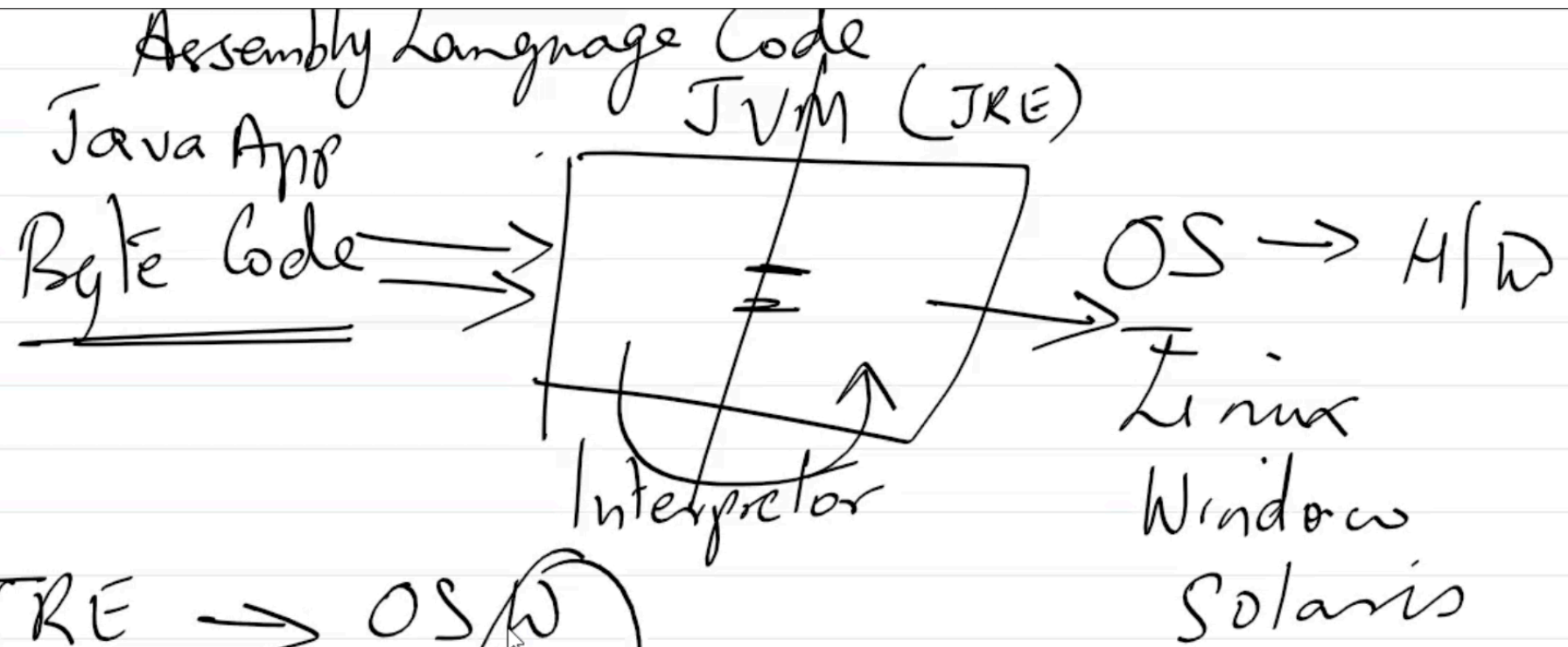
76%

18:39

08-09-2026







JRE  $\rightarrow$  OS

(W, L, S)

① J2SE

New.

JSE

Java Standard  
Edition

② J2E EE

Java Enterprise  
Ed.

③ J2ME

Java  
Micro  
Ed



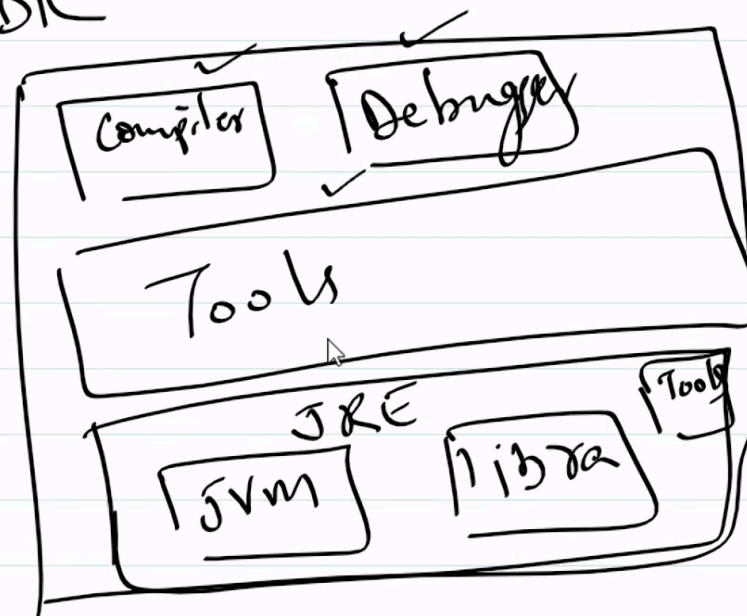
17-0

Java SE

✓ a) JDK

→ b) JRE

JDK



Search



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→ ~~C:\javase~~  
D:\javase

100% oop

- ① everything shd be an object
- ② nothing shd be global
- ③ All features of OOPs to be supported



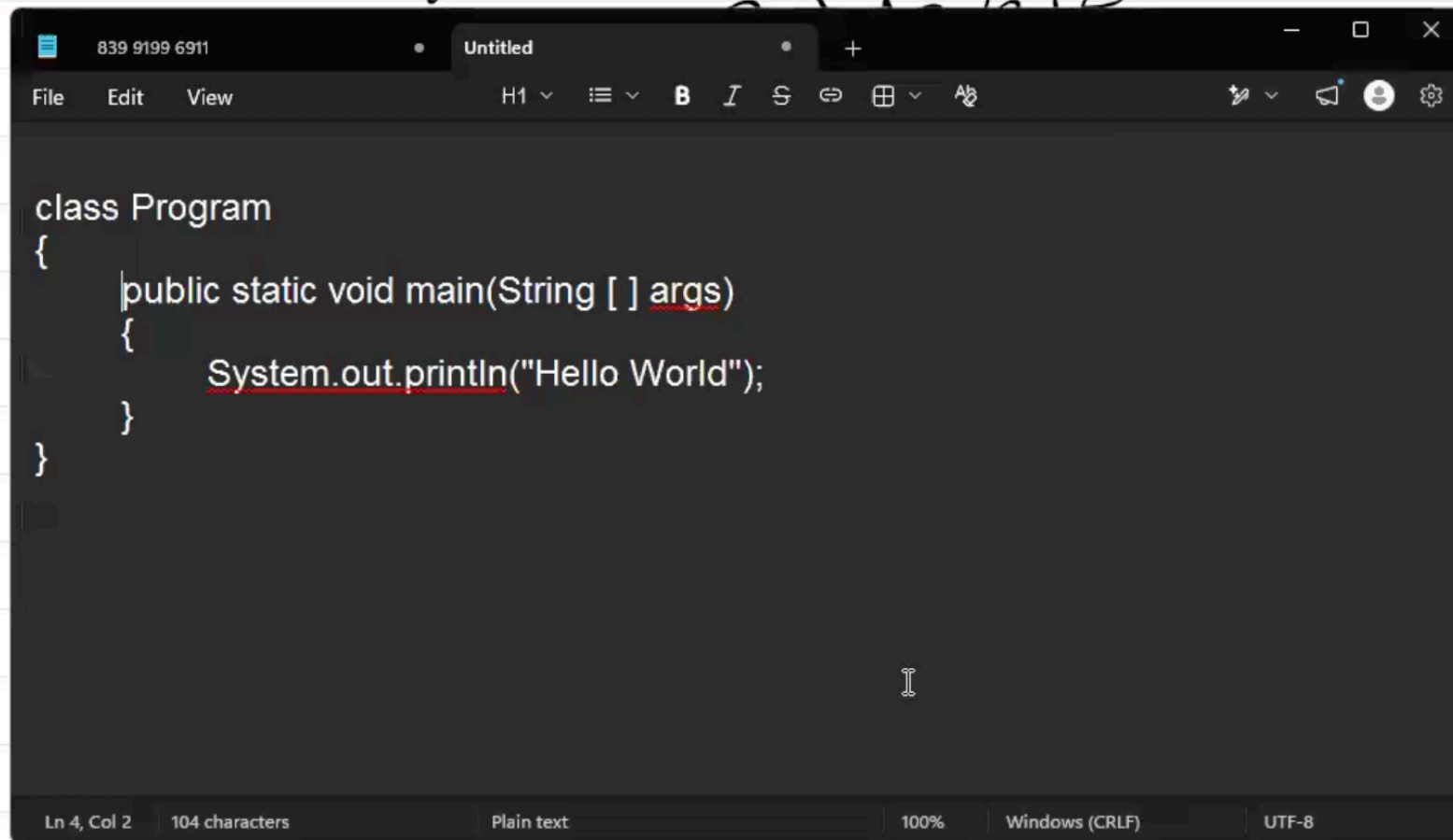
Search



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Talking: P Jagasia



A screenshot of a code editor window with a dark theme. The window has a title bar with '839 9199 6911' and 'Untitled'. The menu bar includes 'File', 'Edit', and 'View'. The toolbar shows various icons for text formatting and editing. The code is written in Java and is as follows:

```
class Program
{
    public static void main(String [ ] args)
    {
        System.out.println("Hello World");
    }
}
```

The status bar at the bottom displays 'Ln 4, Col 2', '104 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'. A handwritten arrow points from the top of the page to the code editor window.

```
E:\kharghar>javac -version  
javac 22
```

```
E:\kharghar>javac Anything.java
```

```
E:\kharghar>dir  
Volume in drive E is E  
Volume Serial Number is 1889-316D
```

```
Directory of E:\kharghar
```

```
08-09-2026 19:16 <DIR> .  
08-09-2026 19:13      111 Anything.java  
08-09-2026 19:16      420 Program.class  
                2 File(s)      531 bytes  
                1 Dir(s)  144,278,634,496 bytes free
```

```
E:\kharghar>java Program  
Hello World
```

```
E:\kharghar>|
```